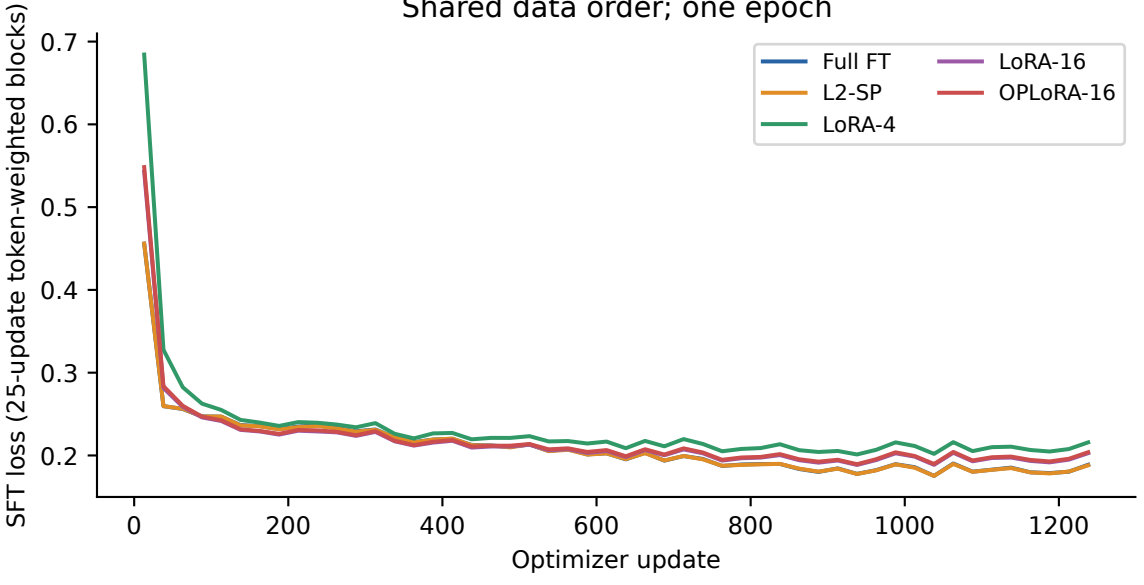
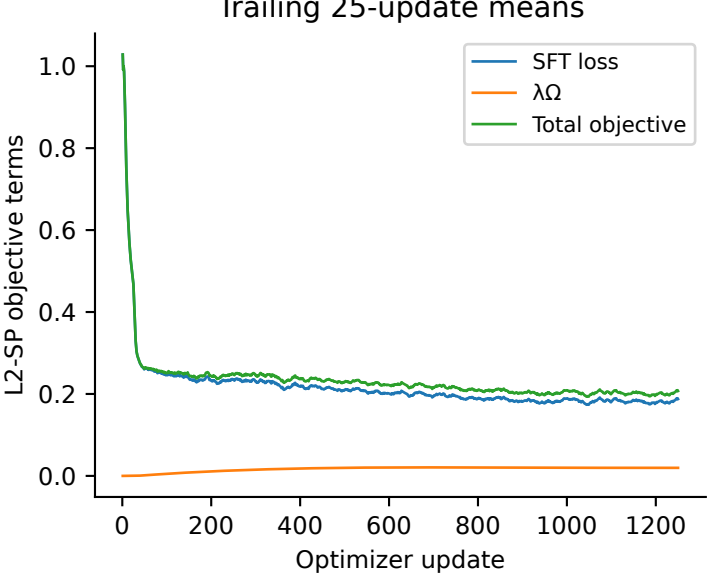

Shared data order; one epoch
SFT loss (25-update token-weighted blocks)
Optimizer update
Full FT
L2-SP
LoRA-4
LoRA-16
OPLoRA-16
0.7
0.6
0.5
0.4
0.3
0.2
0
200
400
600
800
1000
1200


Trailing 25-update means
L2-SP objective terms
Optimizer update
SFT loss
λΩ
Total objective
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200